

Evaluating Causal Reasoning Chains in Multimodal Large Language Models for Brain Imaging

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Abstract

Current neuroimaging evaluations of MLLMs overwhelmingly focus on final-answer correctness, obscuring a critical visual-to-clinical reasoning gap in which models exploit superficial correlations rather than clinically valid logic. Trustworthy diagnosis requires a structured reasoning chain from modality identification through lesion characterization to diagnostic synthesis and treatment planning, a progression that flat benchmarks cannot assess. We propose a causal-chain evaluation framework that transforms flat VQA structures into directed acyclic graphs, enforcing sequential reasoning across five clinical phases via per-node gating. Building on OmniBrainBench’s 9,527 clinically verified VQA pairs across 15 imaging modalities, we will evaluate state-of-the-art MLLMs using a mechanistic failure taxonomy that distinguishes between perception, reasoning, and decision-making failures. We hypothesise that current models will exhibit systematic, modality-dependent degradation in diagnostic and prognostic reasoning, with structural limitations invisible to flat benchmarks. If validated, this framework would establish a new standard for trustworthy medical AI evaluation, extensible beyond neuroimaging to other high-stakes diagnostic domains.

1 Problem Statement

1.1 Background and Motivation

Brain imaging diagnosis requires clinicians to integrate findings from structural MRI, diffusion-weighted imaging, fMRI, and molecular modalities such as PET and SPECT using a structured, sequential workflow. Radiologists progress from anatomical interpretation and lesion characterisation through diagnostic synthesis to prognostic reasoning and treatment planning, with each step causally dependent on the accuracy of the preceding one.

MLLMs have emerged as promising tools for automating this workflow, performing well on Med-VQA benchmarks such as VQA-RAD, SLAKE, and OmniMedVQA. However, these benchmarks evaluate questions in isolation and score only final-answer correctness, which is clinically insufficient for two reasons. First, models can achieve high accuracy through shortcut learning, exploiting dataset biases or memorised pretraining patterns rather than grounding answers in clinically meaningful visual features, a phenomenon well documented in radiology AI [7]. Second, flat benchmarks cannot reveal where reasoning fails, making it impossible to distinguish between a model that identifies a lesion but errs in grading and one that cannot recognise basic anatomy, two failures with fundamentally different clinical safety implications.

1.2 The Visual-to-Clinical Reasoning Gap

We define the visual-to-clinical reasoning gap as the disconnect between producing correct answers and reaching them through valid stepwise clinical reasoning. OmniBrainBench [1] quantifies this gap: the top proprietary model, Gemini-2.5-Pro, achieves only 66.58

GEMeX [2] further shows that models like GPT-4o-mini frequently substitute shortcut reasoning for genuine visual grounding, retrieving plausible answers from pretraining memory rather than reasoning from image content. In clinical settings, such answers are indistinguishable from genuinely reasoned ones when evaluated solely on correctness.

1.3 Research Gap and Proposed Contribution

No existing benchmark enforces the sequential, causally dependent reasoning that characterizes real clinical workflows. OmniBrainBench, NOVA, and Brain Tumour VQA all report per-phase accuracy independently, making it impossible to determine whether a correct diagnostic answer reflects genuine perceptual grounding or shortcut reasoning.

Consider a glioblastoma case: a surgical candidacy question depends causally on inferred tumour aggressiveness and lesion location, which in turn depend on correctly identified enhancement patterns and lesion boundaries. Encoding these dependencies as directed edges in a causal graph prevents models from bypassing earlier reasoning steps, exposing shortcut behaviour that flat benchmarks cannot detect.

We propose the first causal-chain evaluation framework for MLLMs in neuroimaging, transforming OmniBrainBench’s flat VQA pairs into directed acyclic causal graphs and enforcing sequential evaluation via per-node gating. This enables, for the first time, a mechanistic distinction between failures in perception, reasoning, and decision-making in brain-imaging AI.

1.4 Research Questions and Hypotheses

This project is organised around three concrete research questions:

- RQ1:** Do MLLMs show systematic performance degradation as causal chain depth increases, distinguishable from isolated node-level failures?
- RQ2:** Is this degradation modality-dependent, differing between structural modalities such as T1CE and FLAIR and functional modalities such as fMRI and PET?
- RQ3:** Are early perceptual failures causally predictive of downstream reasoning failures, or do models compensate through memorized clinical knowledge?

We hypothesize that chain-completion accuracy will drop by over 20 points relative to node-level accuracy in clinical phases 3

to 5, with the steepest degradation at the perception-to-reasoning transition for ambiguous modalities such as FLAIR and T1CE. We further hypothesize that medical-domain MLLMs will outperform general-purpose models in early perceptual phases but not in later prognostic and therapeutic phases requiring clinical knowledge integration.

2 Related Works

2.1 Medical Visual Question Answering Benchmarks

Medical VQA benchmarks have grown progressively in scale and scope. VQA-RAD [3] established the foundational paradigm of clinically generated radiology QA pairs, SLAKE [4] added semantic labels and knowledge enhancement, and OmniMedVQA [5] expanded modality and body-part coverage to encourage generalization. Despite this progression, all three treat questions as isolated items assessed only on final-answer correctness, ignoring sequential clinical dependencies and making no distinction between genuine reasoning and shortcut learning. Our work addresses this by introducing sequential causal dependencies as a primary evaluation criterion.

2.2 Explainability in Medical VQA

GEMeX [2] is the largest chest X-ray VQA dataset to date and the first to provide both textual reasoning and visual grounding annotations per QA pair. Its key finding, that models such as GPT-4o-mini rely on pretraining memory rather than genuine visual grounding, directly motivates our work. However, GEMeX does not enforce sequential reasoning between questions, leaving clinical causal structure unevaluated. Our framework extends this by making causal chain integrity a measurable criterion.

2.3 Brain Imaging Benchmarks

OmniBrainBench [1] is the most comprehensive brain imaging VQA benchmark, comprising 9,527 clinically verified QA pairs across 15 modalities and five clinical phases. Its comparison of 24 MLLMs with human clinicians reveals persistent deficiencies in complex reasoning tasks, which directly motivates our work. Critically, it reports per-phase accuracy independently, making it impossible to determine whether correct diagnostic answers reflect genuine perceptual grounding. Our framework builds on OmniBrainBench by transforming its flat QA pairs into causally ordered evaluation graphs.

Earlier benchmarks such as Brain Tumour VQA [10] and NOVA [9] confirm the broader pattern: evaluation remains fragmented across isolated tasks rather than organized around the full clinical continuum.

2.4 Shortcut Learning and Reasoning Failures in Clinical AI

DeGrave et al[7] demonstrated that radiology AI models achieve high accuracy by exploiting confounding factors rather than true pathological features, establishing shortcut learning as a systemic problem and motivating evaluation frameworks that verify feature validity, not just correctness. Wei et al[8] further showed

that without explicit chain-of-thought prompting, LLMs default to single-step answer generation that bypasses intermediate reasoning entirely. In clinical settings where intermediate steps carry independent diagnostic significance, our causal chain framework directly targets this failure mode.

3 Technical Approach

3.1 Overview

Our technical approach has three components: causal graph construction, which transforms OmniBrainBench’s flat VQA pairs into directed acyclic graphs that encode clinical dependencies; a soft-gated evaluation framework that enforces causal ordering while preserving complete evaluation data across all models; and a mechanistic failure taxonomy that decomposes errors into perception, reasoning, and decision-making failures. Together, these form a model-agnostic pipeline applicable to any MLLM without architectural modification.

3.2 Causal Graph Construction

The first component transforms OmniBrainBench’s 9,527 flat VQA pairs into structured causal graphs, where each graph corresponds to a single clinical case and directed edges encode reasoning dependencies between questions.

We adopt OmniBrainBench’s five clinical phases directly as the structural backbone of our causal graphs [1]: Anatomical and Imaging Assessment (AIA), Lesion Identification and Localization (LIL), Diagnostic Synthesis and Causal Reasoning (DSCR), Prognostic Judgment and Risk Forecasting (PJRF), and Therapeutic Cycle Management (TCM).

Within this structure, questions are further organized into three cognitive layers. The Perception layer covers direct visual interpretation tasks, primarily AIA and LIL. The Reasoning layer covers inferential integration of perceptual observations, primarily DSCR. The Decision layer covers high-level clinical judgment requiring prior reasoning, corresponding to PJRF and TCM.

This mapping is initially rule-based, using OmniBrainBench’s validated task labels, with a GPT-based classification step to resolve ambiguous cases and radiologist validation of a sample of graph assignments before the benchmark is finalized.

3.2.1 Encoding Causal Dependencies. Directed edges encode clinical dependencies between nodes, where an edge from A to B indicates that a correct answer to B depends on a correct answer to A. For example, a surgical candidacy node depends on parent nodes for lesion location and inferred tumour aggressiveness. Within-layer edges are rare because perception nodes feed into reasoning nodes jointly rather than depending on each other sequentially. Cross-layer edges are mandatory: every reasoning node must have at least one perception parent, and every decision node at least one reasoning parent, enforcing the clinical logic that no high-level conclusion is valid without demonstrated lower-level grounding.

3.3 Soft-Gated Sequential Evaluation Framework

We adopt soft gating rather than hard gating for a principled reason: hard gating blocks downstream evaluation when upstream nodes

fail, leading to missing data that renders cross-model comparisons statistically unreliable. Soft gating instead evaluates every node for every model, propagating upstream performance as a multiplicative weight on downstream scores, preserving complete evaluation data while capturing causal dependency structure.

3.3.1 Node-Level Scoring. Each node is scored independently using the model’s response to the corresponding VQA question. Multiple-choice questions are scored as binary correct or incorrect. Open-ended questions are scored using BERTScore against the ground-truth answer, consistent with GEMeX [2].

3.3.2 Soft Gating and Weighted Chain Scores. For each node i with a set of parent nodes $P(i)$ in the causal graph, the gated score $\hat{s}_i$ is computed as:

$$\hat{s}_i = s_i \times \frac{1}{|P(i)|} \sum_{j \in P(i)} \hat{s}_j \quad (1)$$

This formulation means a node’s effective score is its raw accuracy weighted by the average gated score of its parent nodes. A model that correctly answers a diagnostic question but has failed all upstream perception questions receives a heavily discounted gated score, reflecting the clinical reality that an accidentally correct diagnosis built on incorrect perception is less trustworthy than one grounded in accurate observation. A model that succeeds at both perception and diagnosis receives the full score at the diagnostic node. For root nodes (perception-layer nodes with no parents), the gated score equals the raw score: $\hat{s}_i = s_i$.

3.3.3 Visual Justification Evaluation. Answer correctness alone is insufficient for reasoning and decision nodes, since models may produce correct answers through pretraining memory rather than genuine visual grounding [2, 7]. We therefore require models to justify reasoning and decision node responses by explicitly referencing visual features identified in earlier perception nodes. Justifications are evaluated using semantic similarity against ground-truth clinical rationales, with hallucinations flagged when stated visual features are inconsistent with OmniBrainBench’s clinical annotations. This grounding verification step distinguishes a model that reaches the correct diagnosis for the right visual reasons from one relying on spurious knowledge retrieval.

3.3.4 Chain-Level and Phase-Level Metrics. We derive three aggregate metrics from node-level gated scores. The chain completion score is the mean gated score across all nodes in a case, measuring overall causal reasoning integrity. Phase accuracy is the mean gated score per clinical phase, directly addressing RQ1. The error propagation coefficient measures the conditional probability that a perception node failure predicts downstream reasoning failure, directly addressing RQ3.

Together these metrics allow us to pinpoint exactly where the visual-to-clinical reasoning gap occurs, whether at perception, diagnostic synthesis, or treatment planning, with a precision that flat benchmarks cannot achieve.

3.4 Mechanistic Failure Taxonomy

We classify model errors into three categories. A perception failure occurs when a model fails at a Perception-layer node, such as misidentifying ring enhancement or imaging modality, invalidating

all downstream reasoning. A reasoning failure occurs when perception nodes are correct but a Reasoning-layer node fails, invisible to flat benchmarks since models may still produce correct answers through shortcut retrieval. A decision-making failure occurs when both perception and reasoning nodes are correct but a Decision-layer node fails, the highest-stakes failure mode in terms of patient safety.

Formally, a perception failure is identified when $s_i < \theta$ at a Perception node, a reasoning failure when $s_i < \theta$ at a Reasoning node but all parent gated scores $\hat{s}_j \geq \theta$, and a decision-making failure when $s_i < \theta$ at a Decision node but all upstream gated scores are at or above threshold. We propose $\theta = 0.5$ with sensitivity analysis across $\theta \in \{0.4, 0.5, 0.6\}$.

3.5 Model-Agnostic Pipeline Design

The evaluation pipeline accepts a causal graph and associated imaging data, presents each node’s question in topological order, records responses and visual justifications, applies soft gating, and classifies failures according to the three-way taxonomy. This design ensures uniform application across open-source models such as LLaVA-Med and InternVL3, medical-domain models such as HuatuoGPT-Vision and Lingshu, and proprietary models such as GPT-5 and Gemini-2.5-Pro, without requiring access to model internals.

4 Experiments

4.1 Overview

Our experimental design comprises three planned experiments corresponding directly to the research questions in Section ?? . Experiment 1 benchmarks a representative set of MLLMs on the full causal chain evaluation suite, addressing RQ1. Experiment 2 stratifies results across imaging modality groups to identify modality-dependent degradation profiles, addressing RQ2. Experiment 3 computes error propagation coefficients across adjacent node pairs to determine whether perceptual failures causally predict downstream reasoning failures, addressing RQ3.

4.2 Dataset

All experiments use a stratified subset of OmniBrainBench [1], comprising 9,527 clinically verified VQA pairs across 15 brain imaging modalities and five clinical phases. Because the task distribution is heavily skewed, Disease Diagnosis Reasoning alone accounts for 5,076 pairs while tasks such as Drug Response Prediction contain as few as 10, we apply stratified sampling of up to 200 pairs per task, yielding a working set of approximately 2,000 - 3,000 pairs that ensures all phases contribute equally to the evaluation.

The physician baseline of 91.35% reported by OmniBrainBench serves as our human reference point. We acknowledge this baseline was established on flat VQA rather than causal chain format, and identifying its causal chain equivalent is reserved for future work.

4.3 Model Selection

4.4 Model Selection

Rather than replicating OmniBrainBench’s full 24-model evaluation, we select 12 representative models spanning the full performance range within each category. We include the top two and bottom one

performers per category from OmniBrainBench, plus LLaVA-Med as a historical medical VQA baseline.

From the proprietary category: Gemini-2.5-Pro (66.58%), GPT-5, and DeepSeek-V3.1 (40.14%). From the medical-domain category: HuatuoGPT-Vision-34B (63.56%), Lingshu-32B (54.39%), and LLaVA-Med-7B (38.84%). From the open-source category: Qwen3-VL-30B (56.40%), InternVL3-38B (44.38%), and Janus-Pro-7B (45.11%). This selection ensures cross-category patterns are not artefacts of uneven representation while remaining computationally tractable.

4.5 Evaluation Metrics

Performance is measured at three levels of granularity. *Node-level accuracy* measures correctness at each discrete reasoning step. *Phase-level accuracy* measures the mean gated score across all nodes in a clinical phase. *Chain-completion rate* measures the proportion of cases where a model navigates the full causal graph above threshold, the most stringent test of end-to-end reasoning integrity. We additionally report *visual grounding rate*, the proportion of reasoning and decision node responses with valid visual justifications, and *failure-type distribution* across the three taxonomy categories.

4.6 Experiment 1: Causal Chain Benchmark

Each model is evaluated on the full causal chain suite in topological node order, with soft gating applied, visual justifications assessed, and failures classified according to the three-way taxonomy.

We project that all models will show declining performance as chain depth increases, consistent with Gemini-2.5-Pro dropping from 66.58% overall to 40.84% on risk stratification in OmniBrainBench [1]. The gap between flat VQA accuracy and gated chain accuracy is expected to be largest for proprietary models that achieve high scores through shortcut reasoning [2], whose downstream scores the gating mechanism will discount when perception nodes fail.

4.7 Experiment 2: Modality-Stratified Analysis

Experiment 2 stratifies Experiment 1 results across three modality groups, a novel contribution extending beyond OmniBrainBench’s aggregate reporting. The structural group covers T1W, T1CE, T2W, FLAIR, and PD. The diffusion and functional group covers DWI, SWI, MRA, and fMRI. The molecular group covers PET and SPECT.

We project that degradation will be most severe for functional and molecular modalities, where visual interpretation requires domain-specific expertise underrepresented in general-purpose pretraining corpora, and that medical-domain models will show a smaller performance gap across modality groups than general-purpose models.

4.8 Experiment 3: Error Propagation Analysis

For every adjacent node pair across all causal graphs and models, we compute the conditional probability of a child node failing given a parent node failure and compare it against the unconditional child failure rate. A high error propagation coefficient indicates perceptual failures cascade through the reasoning chain; a low coefficient indicates models compensate through memorised clinical knowledge despite incorrect perceptual grounding.

This distinction is clinically significant: a model whose errors cascade from perception is limited by its visual encoder, while one whose errors arise at the synthesis stage despite correct perception is limited by its clinical reasoning capacity. Our framework is the first to make this distinction measurable in neuroimaging.

We project the coefficient will be high for open-source and smaller medical-domain models and lower for large proprietary models such as GPT-5 and Gemini-2.5-Pro. If confirmed, this would suggest that high-performing proprietary models achieve their benchmark scores through a fundamentally different mechanism than the grounded visual reasoning safe clinical deployment requires, making flat benchmark performance a poor predictor of real-world reliability.

5 Conclusion

5.1 Summary of Proposed Work

We propose the first causal-chain evaluation framework for MLLMs in brain imaging, enforcing sequential clinically grounded reasoning as a measurable criterion rather than treating diagnostic accuracy as an isolated endpoint. Building on OmniBrainBench’s 9,527 clinically verified VQA pairs across 15 imaging modalities, we will construct directed acyclic causal graphs encoding clinical dependencies and evaluate models using a soft-gated scoring mechanism that rewards genuine perceptual grounding and penalises shortcut reasoning, supported by a three-way failure taxonomy distinguishing perception, reasoning, and decision-making failures.

5.2 Expected Contributions

This work makes four concrete contributions. First, a causal-chain evaluation suite transforming OmniBrainBench’s flat VQA structure into causally ordered DAGs, the first brain imaging benchmark to enforce sequential reasoning dependencies as a formal evaluation constraint. Second, a soft-gated scoring framework that preserves complete evaluation profiles across all models and nodes, enabling statistically robust cross-model comparison unlike hard-gating approaches that produce missing data. Third, a mechanistic failure taxonomy distinguishing perception, reasoning, and decision-making failures, directly actionable for model developers since each failure type requires a different architectural intervention. Fourth, a modality-stratified degradation analysis providing the first systematic evidence of modality-dependent reasoning degradation in brain imaging MLLMs, informing which modalities require targeted domain adaptation before clinical deployment.

5.3 Broader Significance

The visual-to-clinical reasoning gap our framework addresses is not unique to neuroimaging; it is a general property of any medical domain requiring structured multi-step clinical inference, making our approach applicable in principle to cardiology, oncology, and ophthalmology. Beyond benchmarking, our framework has direct regulatory implications: current medical AI evaluation focuses on endpoint sensitivity and specificity, the same flat accuracy paradigm our work identifies as insufficient. A model achieving high endpoint accuracy through shortcut retrieval may pass regulatory review while remaining clinically unsafe. Ultimately, this project advances

trustworthy AI by ensuring model outputs are not only correct but arrived at through clinically coherent and safe pathways.

5.4 Limitations and Future Work

Three limitations define directions for future work. First, the physician baseline was established on flat VQA rather than causal chain format; future work should establish a dedicated human baseline on the causal chain benchmark. Second, causal graph construction combines rule-based mapping with GPT-based verification, introducing potential annotation errors; future work should conduct an inter-rater reliability study with multiple clinical reviewers. Third, the soft gating formula uses a simple multiplicative weighting scheme; future work could explore learned weights that better capture asymmetric clinical dependencies in complex cases such as multi-lesion presentations or rare disease subtypes.

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